Piecewise Rules for Rates and Fees
Answer Key

Algebra 1

Quick Answers

1. 13
2. 11
3. 103
4. 7
5. 3.5
6. 78
7. 10
8. —
9. $t = 4$, $y = 10$
10. 0.16

Curriculum

Level: Algebra 1
HSA-CED.A.1–A.4 – *problems 2, 3, 4, 5, 9, 10*
HSF-IF.A–HSF-IF.C – *problems 1, 6, 7, 8*
Difficulty 1–5 of 5 across 10 tagged checks.

Common wrong answers

3. *If they got 128.0: billed every kilowatt-hour at the second-tier rate instead of only the ones above the 500 kWh allowance*

1. City garage: $4.00 up to 2 hours, $3.00 per additional hour, up to 8 hours. Find $C(5)$.

The flat fee covers everything up to the breakpoint at $t = 2$; past it, only the hours *beyond* the first 2 are charged, which is $t - 2$ of them.

$$C(t) = \begin{cases} 4.00, & 0 < t \leq 2 \\ 4.00 + 3.00(t-2), & 2 < t \leq 8 \end{cases}$$

A 5-hour stay satisfies $2 < 5 \leq 8$, so use the second branch: $C(5) = 4.00 + 3.00(5-2) = 4.00 + 3.00 \cdot 3 = 4.00 + 9.00$ dollars. $\boxed{C(5) = 13.00}$

2. Shipper: $6.00 up to 3 pounds, then $1.25 per pound over 3. Find $C(7)$.

Same shape, different units. The charged quantity past the breakpoint is $w - 3$ pounds.

$$C(w) = \begin{cases} 6.00, & 0 < w \leq 3 \\ 6.00 + 1.25(w-3), & w > 3 \end{cases}$$

A 7-pound package uses the second branch: $7 - 3 = 4$ pounds are charged at $1.25 each, so $C(7) = 6.00 + 1.25 \cdot 4 = 6.00 + 5.00$ dollars. $\boxed{C(7) = 11.00}$

3. Utility: 500 kilowatt-hours at \$0.11, everything above 500 at \$0.16. Priya used 800 and claims \$128.00.

$$C(n) = \begin{cases} 0.11n, & 0 \le n \le 500 \\ 0.11(500) + 0.16(n - 500), & n > 500 \end{cases}$$

Priya's mistake: she applied the second-tier rate to the whole month, computing $0.16 \times 800 = 128.00$. The higher rate applies only to the kilowatt-hours *above* the 500 allowance; the first 500 always bill at \$0.11.

Correct bill: the first 500 kilowatt-hours cost $0.11 \times 500 = 55.00$ dollars. The remaining $800 - 500 = 300$ kilowatt-hours cost $0.16 \times 300 = 48.00$ dollars. Total: $55.00 + 48.00$ dollars. $\boxed{C(800) = 103.00}$

4. Taxi: \$3.50 for the first mile, \$2.25 per additional mile. Find the trip length whose fare is \$17.00.

$$C(m) = \begin{cases} 3.50, & 0 < m \le 1 \\ 3.50 + 2.25(m - 1), & m > 1 \end{cases}$$

A \$17.00 fare is well past the \$3.50 minimum, so it lives on the second branch. Set that branch equal to the fare and solve:

$$\begin{aligned} 3.50 + 2.25(m - 1) &= 17.00 \\ 2.25(m - 1) &= 13.50 \\ m - 1 &= 6 \\ m &= 7 \end{aligned}$$

Check: $3.50 + 2.25 \cdot 6 = 3.50 + 13.50 = 17.00$ dollars. The trip was 7 miles. $\boxed{m = 7 \text{ mi}}$

5. Pool: \$8.00 for the first 3 hours, k dollars per additional hour; a 7-hour visit costs \$22.00. Find k.

$$C(t) = \begin{cases} 8.00, & 0 < t \le 3 \\ 8.00 + k(t - 3), & t > 3 \end{cases}$$

A 7-hour visit is charged for $7 - 3 = 4$ extra hours:

$$\begin{aligned} 8.00 + k(7 - 3) &= 22.00 \\ 4k &= 14.00 \\ k &= 3.50 \end{aligned}$$

Each hour past the first 3 costs \$3.50, so the rule is $C(t) = 8.00 + 3.50(t - 3)$ for $t > 3$. $\boxed{k = 3.50}$

6. Phone plan: \$30.00 base with 5 gigabytes included, \$12.00 per gigabyte above 5, bill capped at \$90.00. Find $C(9)$.

Three regions, so three branches — the cap is a branch of its own:

$$C(g) = \begin{cases} 30.00, & 0 \le g \le 5 \\ 30.00 + 12.00(g-5), & 5 < g \le 10 \\ 90.00, & g > 10 \end{cases}$$

At $g = 9$ the middle branch applies, and $9 - 5 = 4$ gigabytes are charged: $C(9) = 30.00 + 12.00 \cdot 4 = 30.00 + 48.00$ dollars, which is under the \$90.00 cap, so the cap does not bind.

$$\boxed{C(9) = 78.00}$$

7. Same plan: find the usage at which the middle branch first reaches the \$90.00 cap.

The cap takes over exactly where the middle branch's value hits \$90.00:

$$\begin{aligned} 30.00 + 12.00(g-5) &= 90.00 \\ 12.00(g-5) &= 60.00 \\ g - 5 &= 5 \\ g &= 10 \end{aligned}$$

So the middle branch runs on $5 < g \le 10$ and the flat \$90.00 branch takes over for $g > 10$.

$$\boxed{g = 10 \text{ GB}}$$

8. Explain why the middle branch and the capped branch must agree where they meet. *(Open response — flagged for manual review.)*

Model answer: At $g = 10$ the middle branch gives $30.00+12.00(10-5) = 90.00$ dollars and the capped branch gives \$90.00. They agree, so the graph of C is one connected path with no vertical break. If they disagreed — say the cap were \$85.00 instead — then a customer using exactly 10 gigabytes would pay \$90.00 while a customer using 10.1 gigabytes would pay \$85.00. Using *more* data would lower the bill, so a customer could save money by streaming a video they did not want, and two customers whose usage differs by a rounding error would see bills \$5.00 apart. A fee schedule is only fair if cost never jumps down as usage goes up, which is exactly the requirement that the branches match at the breakpoint.

Grading note: full credit for any answer that (a) evaluates both branches at the breakpoint and observes they agree, and (b) describes the jump a customer would experience otherwise. Naming it "continuity" is welcome but not required at this level.

9. Garage A (\$4.00 up to 2 hours, then \$3.00 per hour) versus Garage B (\$2.50 per hour). Find the stay longer than 2 hours on which they cost the same.

For $t > 2$, Garage A costs $4.00 + 3.00(t - 2)$ and Garage B costs $2.50t$. Let y be the shared cost in dollars:

$$\begin{cases} y = 4.00 + 3.00(t - 2) \\ y = 2.50t \end{cases}$$

Substituting the second equation into the first:

$$\begin{aligned} 2.50t &= 4.00 + 3.00(t - 2) \\ 2.50t &= 3.00t - 2.00 \\ 2.00 &= 0.50t \\ t &= 4 \end{aligned}$$

Then $y = 2.50 \cdot 4 = 10.00$ dollars, and the check on Garage A gives $4.00 + 3.00(4 - 2) = 10.00$ dollars as well. The answer $t = 4$ lies in the $t > 2$ region the system was written for, so it is a legitimate solution. $\boxed{t = 4,\ y = 10.00}$

10. Print shop: \$25.00 setup plus \$0.40 per page for the first 200 pages, then r dollars per page beyond 200; a 500-page job costs \$153.00. Find r.

$$C(p) = \begin{cases} 25.00 + 0.40p, & 0 \le p \le 200 \\ 25.00 + 0.40(200) + r(p - 200), & p > 200 \end{cases}$$

The first 200 pages always cost $25.00 + 0.40 \times 200 = 105.00$ dollars, and a 500-page job has $500 - 200 = 300$ pages at the reduced rate:

$$\begin{aligned} 105.00 + r(500 - 200) &= 153.00 \\ 300r &= 48.00 \\ r &= 0.16 \end{aligned}$$

So the reduced rate is \$0.16 per page and the rule is $C(p) = 105.00 + 0.16(p - 200)$ for $p > 200$. $\boxed{r = 0.16}$